

NGUYỄN GIA NGHI

Ho Chi Minh City, Vietnam | gianghi30032005@gmail.com | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

RESEARCH INTERESTS

Computer systems performance | Distributed systems and coordination | Software engineering | Latency diagnosis

EDUCATION

VNUHCM-University of Science

Sep 2023–May 2027

Bachelor of Science | Major: Information Technology

Ho Chi Minh City

GPA: 8.34/10.00 (3.58/4.00) | 121 credits completed

Expected thesis completion: March 2027

UNDERGRADUATE RESEARCH THESIS

Latency diagnosis and optimization mechanisms for edge cloud-gaming systems | In progress

Supervisor: Professor Lê Ngọc Sơn

Research title (English translation): Latency diagnosis and optimization mechanisms for edge cloud-gaming systems

Official title (Vietnamese): Nghiên cứu cơ chế chẩn đoán và tối ưu hóa độ trễ cho hệ thống trò chơi đám mây biên

- Studying whether latency fingerprints, stored as context-specific patterns of how the system responds to bounded relief actions, can help distinguish cross-layer bottlenecks and guide more targeted recovery than generic adaptation across heterogeneous edge-cloud-gaming environments.
- Built a working, self-hosted edge-cloud-gaming testbed with Dockerized game workloads, desktop host orchestration, GStreamer capture and encoding, WebRTC media and input paths, a lightweight browser client, and cross-layer telemetry.
- Implemented the first end-to-end fingerprinting slice, including versioned records, response-vector construction, fingerprint storage, context filtering, and evidence-backed decisions. A separately collected run matched the initial seed with a provisional match strength of 0.982 across 22 shared features, providing an early repeatability check.
- Planning controlled comparisons with fixed-profile, threshold-based, and passive-telemetry baselines, evaluated through diagnosis accuracy, P95/P99 latency, frame-deadline misses, recovery time, quality degradation, and probe overhead. A secondary comparison will test the controller with and without deadline-aware stale-work handling.

RELEVANT ACADEMIC PREPARATION

Selected strong results (10-point scale): Discrete Mathematics (9.40) | Introduction to Cloud Computing (9.10) | Computer Systems (9.10) | Software Design (8.50) | Applied Statistics for Engineers and Scientists (8.10) | Operating Systems (8.00) | Introduction to Databases (8.00)

SYSTEMS AND SOFTWARE PROJECTS

Latency Fingerprinting | Python research engine | [Repository](#)

- Developed a versioned Python package and CLI that validates records against versioned schemas, ingests testbed records, constructs response vectors, and matches fingerprints only in compatible contexts.

PIXELATED Studio Edition | Creator platform and self-hosted edge-cloud-gaming research testbed

- Built and released a TypeScript/React creator platform, then extended it into a self-hosted edge-cloud-gaming testbed that streams interactive workloads from a user-owned host to a lightweight browser client.
- Implemented a hosted Fastify control plane, Electron-based host orchestration, a Dockerized runtime, GStreamer capture and encoding, secure pairing and session flows, WebRTC media and input paths, and cross-layer telemetry.

[Repository](#) | [Live application](#) | [Releases](#)

PIXELATED User Edition | Browser-based WebAssembly gaming platform

- Built and deployed a React/PWA application that runs supported NES, Game Boy, and Game Boy Color titles locally in the browser through WebAssembly-based libretro cores, without relying on the Studio desktop engine or WebRTC streaming.
- Instrumented the browser runtime to export launch-time, frame-pacing, long-task, capability, memory, and error telemetry, providing a browser-execution baseline for comparison with the Studio version’s edge-cloud-streaming path.

[Repository](#) | [Live application](#)

INDUSTRY EXPERIENCE

Full-Stack Engineer Intern | TMA Solutions

Jun–Aug 2026
Ho Chi Minh City

- Worked with Jira tickets spanning bug fixes, product improvements, and small features across an AI-assisted project-management application built with MongoDB, Express.js, React, and Node.js.
- Modeled and queried MongoDB data through Mongoose, built API tests, self-tested changes, and actively participated in an Agile/Scrum workflow.

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Java, Python, C/C++, SQL
Frontend	React, Vite, browser Web APIs
Backend	Node.js, Express.js, Fastify, Spring Boot, REST APIs, Socket.IO
Data	PostgreSQL, MySQL, MongoDB
Cloud platforms	AWS, GCP
Infrastructure & DevOps	Linux, Docker, Kubernetes, Helm, Terraform, GitHub Actions
Real-time media	WebRTC, GStreamer
Performance evaluation	Telemetry instrumentation, benchmarking, experimental design, latency analysis

CERTIFICATIONS AND ENGLISH

- AWS Certified Solutions Architect - Associate
- AWS Certified Cloud Practitioner
- IELTS Academic: 8.0 overall - Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 7.0 (test date: 2nd October 2025)